

NALCHIK, Russian Federation

WMO: 372120

Lat: 43.513N

Lon: 43.637E

Elev: 1461

StdP: 13.94

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	9.3	-0.7	5.6	2.8	5.4	7.6	10.1	11.6	35.6	9.3	35.9	1.3	250	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.2	70.4	88.0	69.4	85.8	68.7	73.2	85.4	71.6	84.0	70.2	82.1	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	114.6	81.1	67.7	107.6	78.4	66.1	101.7	76.5	37.8	85.5	36.3	84.5	35.0	82.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	9.5	7.8	DB	-1.1	95.0	8.6	2.5	-7.3	96.8	-12.3	98.2	-17.1	99.6	-23.3	101.4	(4)
			WB	-1.4	75.5	8.3	2.6	-7.4	77.3	-12.2	78.8	-16.8	80.2	-22.8	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	28.6	30.1	40.3	50.8	61.7	69.5	73.7	73.8	64.2	52.3	40.4	33.6
	DBStd	17.67	9.39	10.10	7.08	7.74	6.10	5.24	4.42	5.12	6.63	7.52	7.76	7.17
	HDD50	2493	663	559	311	83	4	1	0	0	2	61	300	509
	HDD65	5654	1128	978	767	428	135	17	1	4	88	396	738	974
	CDD50	3117	0	1	10	107	366	587	735	737	429	133	12	0
	CDD65	804	0	0	0	2	32	153	271	277	66	3	0	0
	CDH74	6994	0	0	2	37	294	1246	2317	2537	533	28	0	0
	CDH80	2355	0	0	0	4	50	360	807	985	146	3	0	0
Wind	WSAvg	3.5	2.6	3.0	3.8	4.2	4.0	4.1	4.0	3.9	3.4	3.0	3.0	2.9
Precipitation	PrecAvg	35.9	2.4	2.3	3.0	3.3	3.9	4.4	3.3	2.8	2.7	2.8	2.5	2.6
	PrecMax	43.8	4.5	4.0	6.0	6.2	7.0	8.8	6.0	5.1	5.7	5.6	4.6	5.8
	PrecMin	28.4	0.7	1.4	1.3	1.5	1.8	1.9	1.0	0.8	0.9	0.8	0.7	1.0
	PrecStd	4.0	0.8	0.7	1.1	1.0	1.1	1.4	1.2	1.0	1.1	1.1	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	60.7	68.3	78.6	85.8	90.0	94.7	95.0	90.0	77.4	67.9	58.6
		MCWB	43.0	46.9	49.1	59.4	66.9	68.8	73.9	70.8	68.0	60.7	51.3	46.6
	2%	DB	46.4	51.7	62.3	73.4	80.6	87.4	91.0	91.5	84.1	71.9	61.3	51.0
		MCWB	39.3	42.9	48.3	56.6	64.5	68.6	71.5	70.1	65.0	58.8	50.3	43.7
	5%	DB	42.6	46.0	57.1	69.6	77.1	84.3	87.7	88.8	80.2	68.2	55.5	46.4
		MCWB	37.6	39.9	46.6	55.6	63.4	68.3	70.3	69.6	64.2	57.7	48.0	40.7
	10%	DB	39.1	41.2	52.0	64.6	73.6	80.9	84.4	85.8	76.7	64.3	51.4	43.0
		MCWB	35.3	37.5	44.5	53.4	61.9	67.2	69.1	69.2	63.3	56.0	46.0	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	48.4	52.6	60.8	69.2	73.8	77.0	74.4	71.7	62.9	54.2	48.0
		MCDB	50.6	58.4	65.3	73.5	81.7	85.8	90.8	86.7	86.3	72.8	61.9	54.2
	2%	WB	40.3	43.9	49.2	58.1	66.4	71.0	73.8	72.7	67.7	60.8	51.4	44.5
		MCDB	44.9	49.7	58.4	70.7	77.6	82.7	85.6	86.0	77.4	69.0	59.2	48.7
	5%	WB	37.7	40.5	47.4	56.3	64.4	69.2	71.9	71.1	65.8	58.8	49.0	42.1
		MCDB	41.3	44.7	55.3	67.0	74.7	80.5	82.9	84.9	76.4	66.2	54.6	45.7
	10%	WB	35.7	37.9	45.2	54.3	62.7	67.7	70.3	70.0	64.4	56.9	46.7	39.4
		MCDB	38.4	40.8	51.8	63.1	72.2	78.3	81.2	82.8	74.2	63.0	50.8	42.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.1	14.3	17.5	17.4	18.0	18.0	19.4	18.0	15.3	12.8	11.7
		MCDBR	17.2	20.1	24.7	25.4	23.5	22.8	22.6	24.1	24.3	23.3	22.5	19.4
		MCWBR	12.0	13.1	13.9	12.5	12.1	10.6	9.5	9.6	10.4	12.8	13.0	13.2
	5% WB	MCDBR	15.7	18.5	22.2	22.2	20.7	20.4	20.1	21.3	20.6	19.9	19.2	17.6
		MCWBR	11.5	12.5	13.2	11.9	11.9	10.8	10.0	9.6	10.0	11.9	12.0	12.5
Clear-Sky Solar Irradiance	taub	0.322	0.358	0.413	0.471	0.442	0.446	0.487	0.484	0.427	0.387	0.342	0.320	
	taud	2.307	2.207	2.108	1.998	2.171	2.212	2.109	2.109	2.239	2.304	2.364	2.326	
	Ebn at Noon	252	259	259	252	265	263	251	246	252	247	242	240	
	Edh at Noon	27	35	44	53	46	44	49	47	38	32	25	24	
All-Sky Solar Radiation	RadAvg	503	756	965	1353	1637	1786	1807	1632	1256	833	556	430	
	RadStd	66	128	102	174	174	114	137	144	140	113	73	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page